


```

> lrtest(probit, probit_restricted)
Likelihood ratio test

Model 1: smoker ~ sex + age + bmi
Model 2: smoker ~ sex
  #Df  LogLik Df Chisq Pr(>Chisq)
1    4 -109.77
2    2 -110.50 -2  1.454      0.4834

# -----
# c)
# marginal effects for the average observation
> (meff = probitmfx(formula=smoker~sex+age+bmi,
+                   data=smoke, atmean=TRUE))
Call:
probitmfx(formula = smoker ~ sex + age + bmi, data = smoke, atmean = TRUE)

Marginal Effects:
      dF/dx  Std. Err.      z    P>|z|
sex -0.2868432  0.0752012 -3.8143 0.0001365 ***
age -0.0018909  0.0026701 -0.7082 0.4788254
bmi -0.0089720  0.0112921 -0.7945 0.4268798
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

dF/dx is for discrete change for the following variables:
[1] "sex"

> # mean marginal effects
> (meff = probitmfx(formula=smoker~sex+age+bmi,
+                   data=smoke, atmean=FALSE))
Call:
probitmfx(formula = smoker ~ sex + age + bmi, data = smoke, atmean = FALSE)

Marginal Effects:
      dF/dx  Std. Err.      z    P>|z|
sex -0.2852462  0.0747266 -3.8172 0.000135 ***
age -0.0017840  0.0025085 -0.7112 0.476972
bmi -0.0084647  0.0105966 -0.7988 0.424400
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

dF/dx is for discrete change for the following variables:
[1] "sex"

# -----
# d)
> options(scipen=999)

```

```
> PseudoR2(probit)
```

McFadden	Adj.McFadden	Cox.Snell	Nagelkerke	McKelvey.Zavoina
0.05850302	0.01561829	0.07624972	0.10272955	0.12241859
Effron	Count	Adj.Count	AIC	Corrected.AIC
0.07937974	0.61046512	0.05633803	227.54129117	227.78081213